Maneesh Kumar Singh

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RESEARCH INTERESTS

Bayesian Scientific Computing, Data Assimilation, Scientific Machine Learning, Numerical Analysis.

EMPLOYMENT

2026– SRM Institute of Science and Technology
Research Assistant Professor, Department of Mathematics
Bayesian data assimilation, Compatible FEM for GFD, Numerical analysis

2022–25 Imperial College London
Postdoctoral Research Associate, Department of Mathematics
Bayesian data assimilation, compatible FEM for GFD

2021–22 University of Konstanz
Postdoc Fellow, Department of Mathematics and Statistics
Space-time FEM for PDEs on moving domains

2019–21 Indian Institute of Science
Postdoc Fellow, Department of Computational and Data Sciences
Operator-splitting FEM, radiative transfer, ParMooN parallel solver

2018 Ashoka University
Teaching Fellow, Department of Mathematics

EDUCATION

2013–19 Ph.D. in Mathematics, Indian Institute of Technology Guwahati

• Thesis title: Analysis of robust computational techniques for singularly perturbed system of parabolic partial differential equations • Supervisor: Prof. Natesan Srinivasan.

2011-13 MSc in Applied Mathematics with 7.86 CGPA, University of Hyderabad

2008-11 BA (Hons) in Mathematics with 68.9 % Banaras Hindu University

FUNDING AND SCHOLARSHIPS

2021 D.S. Kothari Postdoctoral Fellowship, UGC, India

2018 NBHM Postdoctoral fellowship, NBHM, DAE, India

2013 CSIR-UGC JRF, Gate Fellowship in Mathematics, MHRD, India

TEACHING EXPERIENCE

IIT Guwahati – Teaching Assistant

2014– 18 Calculus, Linear Algebra & ODE, Scientific Computing, PDE, Computational Finance

Ashoka University – Teaching Fellow

2018 Calculus

Imperial College London – Substitute Instructor

2022–25 Computational Linear Algebra, Finite Elements

SRM University – Course Instructor

2026– Probability and Queuing theory, Numerical Methods and Analysis

CURRICULUM DEVELOPMENT

SRM University – Course Design

2026 **Ph.D.** Scientific Computing (Modules I–II): Krylov solvers, numerical ODEs, FDM/FVM/FEM for PDEs, integral equations, wavelets

2026 **B.Tech** Computational Mathematics Lab 1 (26MAC3002L): Python, numerical linear algebra, optimisation, regression, PCA, SVM, Monte Carlo

2026 **B.Tech** Computational Mathematics Lab 2 (26MAC3003L): numerical methods, ODE solvers, cryptography and number theory, quantum computing basics

PUBLICATIONS

Journal Articles

Working Articles

1 M. K. Singh: Consistent CutPINNs for the heat equation on curved level-set domains.

2 M. K. Singh, C.J. Cotter, D. Crisan: Data Assimilation for a stochastic Euler equation using global nudged particle filter.

3 S. Frei, M. K. Singh: Eulerian time-stepping method for transient transport problems on moving domains.

Submitted Articles

2026 M. K. Singh, C.J. Cotter: Energy conserving compatible space-time FEM for deterministic and stochastic rotating shallow water. *J. Comput. Phys (under review) arXiv:2610.13624.*

2026 M. K. Singh: Consistent CutPINNs for elliptic PDEs on curved level-set domains. *Comput. Math. Appl. (under review) arXiv 2605.25562.*

2026 M. K. Singh: Consistent CutPINNs for convection-diffusion equations on curved level-set domains. *Adv. Comput. Math. (under review) arXiv:2606.29147.*

Published Articles

2026 M. K. Singh, J Hope-Collins, C.J. Cotter, D. Crisan : Data assimilation using a global Girsanov nudged particle filter. *Statistics and Computing, 36, 251*

2026 D.D. Holm, M.K. Singh, O. D. Street: A comparative numerical study of stochastic Hamiltonian Camassa-Holm equations.
Proceedings of the Royal Society A, 482 (2330), 20250635

2026 D.D. Holm, M.K. Singh: A multiscale Camassa-Holm equation. *Chaos, Solitons and Fractals, 202, 117409*

2025 S. Ganesan, M. K. Singh: Operator-splitting finite element method for solving the radiative transfer equation. *Numerical Algorithms, 98: 1725-1753*

2024 C.J. Cotter, D. Crisan, M. K. Singh: Data assimilation for the stochastic Camassa-Holm equation using particle filtering: a numerical investigation. *Stochastic Transport in Upper Ocean Dynamics III, Mathematics of Planet Earth, 13: 137–160, (Book chapter)*

2024 S. Frei, M. K. Singh: An implicitly extended Crank-Nicolson scheme for the heat equation on a time-dependent domain. *Journal of Scientific Computing, 99(3) 64: 1-29*

2024 M. K. Singh: A discrete-ordinate weak Galerkin method for radiative transfer equation. *Applied Numerical Mathematics, 201: 628–641*

2021 M. K. Singh, G. Singh, S. Natesan: A unified study on superconvergence analysis of Galerkin FEM for singularly perturbed systems of multiscale nature. *Journal of Applied Mathematics and Computing, 66: 221-243*

2020 S. Natesan, M.K. Singh: Robust computational method for singularly perturbed system of parabolic convection-diffusion problems with interior layers. *Computational and Mathematical Methods, 3: e1146*

2020 M. K. Singh, S. Natesan: Numerical solution of 2D singularly perturbed reaction-diffusion system with multiple scales. *Computers and Mathematics with Applications, 80(4): 36-53*

2019 M. K. Singh, S. Natesan: A parameter-uniform hybrid finite difference scheme for singularly perturbed system of parabolic convection-diffusion problems. *International Journal of Computer Mathematics, 97(4): 275-305*

2019 M. K. Singh S. Natesan: Numerical analysis of singularly perturbed system of parabolic convection-diffusion problem with regular boundary layers. *Differential Equations and Dynamical Systems, 30(3):695-717*

2019 M. K. Singh, S. Natesan: A robust computational method for singularly perturbed system of 2D parabolic convection-diffusion problems. *International Journal of Mathematical Modelling and Numerical Optimisation, 9(2): 127-157*

2018 M. K. Singh, S. Natesan: Richardson extrapolation technique for singularly perturbed system of parabolic partial differential equations with exponential boundary layers. *Applied Mathematics and Computation, 333: 254-275*

Conference Proceedings

2024 C. Cotter, D. Crisan, M. K. Singh: The GONG Particle Filter: A Novel Framework for Data Assimilation in Stochastic PDEs, *CASML 2024, IISc Bangalore*

2018 M. K. Singh, S. Natesan: A finite element super convergence approximations for singularly perturbed system of convection-diffusion problems. *AIP Conference Proceedings, FIAM 2018, 1975, 030016*

PRESENTATION

Talks at Conferences(recent)

2026 Nudging Particles Globally: A Girsanov Approach to SPDE Data Assimilation *RAAM–2026, IIT Hyderabad*

2025 A New Paradigm for Data Assimilation: The Global Girsanov Nudged Particle Filter *LACAM–2025, IISER Thiruvananthapuram*

2025 Nudged particle filter : advanced data assimilation framework for SPDEs textit MATHSD–2025, IIITM Gwalior

2025 Sequential data assimilation using nudging particle filter. *PEML2025, London*

2025 Energy-conserving space–time FEM for rotating shallow water. *6th STUOD Annual Workshop (University of Edinburgh, UK)*

2025 Energy-conserving compatible space–time finite element discretisations of the rotating shallow water equations. *Firedrake meeting 2025 (University of Leeds)*

2025 Energy conserving compatible finite element methods for geophysical fluid dynamics *ENUMATH 2025, Heidelberg*

2024 Data Assimilation for Stochastic PDEs: The GONG Particle Filter Framework. *5th STUOD Annual Workshop (INRIA, Renne, France)*

2024 Advanced particle filtering data assimilation framework for Stochastic PDEs. *Firedrake meeting 2024 (University of Oxford)*

2023 Parallelisation schemes for particle filtering data assimilation. *4th STUOD Annual Workshop (IFREMER, France)*

2023 Data assimilation algorithm using advanced particle filters for stochastic PDEs. *Firedrake meeting 2023 (London)*

Invited Talks at University workshop/seminars

2025 Essential ideas in data assimilation *FoCuS-ML, IIITM Gwalior*

2024 Data assimilation with advanced particle filters for stochastic partial differential equations. *TIFR-CAM Seminar Talk*

2024 The Hitchhiker's Guide to FEM: A Beginner's Perspective. *IIIT Gwalior Workshop on PDE*

2023 An implicitly extended Crank-Nicolson scheme for the heat equation on time-dependent domains. *ICL/UCL Numerics Seminar*

2023 Towards nudging particle filter for Stochastic PDEs. *24th STUOD Sandbox Workshop (ICL)*

2021 An operator-splitting finite element method for radiative transfer equation. *Oberseminar Numerik (University of Konstanz)*

2019 Five days workshop on parallel finite element computing using ParMooN. *(IISc)*.

TECHNICAL SKILLS

– Languages: Python (primary), MATLAB, C++

– ML / scientific libraries: NumPy, SciPy, PyTorch, JAX, TensorFlow, ParaView

– FEM & PDE solvers: Firedrake, FEniCS, deal.II, ParMooN

– HPC & parallel computing: PETSc, MPI

– Dev tools: Git, GitHub Actions, Linux/Unix, LaTeX, Jupyter

REFERENCES

1 Prof. Natesan Srinivasan(PhD Supervisor), Department of Mathematics, IIT Guwahati, India
Email: natesan@iitg.ac.in Ph: +91 3612582613

2 Prof. Colin Cotter(Postdoc mentor), Department of Mathematics, Imperial College London, UK
Email: colin.cotter@imperial.ac.uk Ph: +44 2075943468

3 Prof. Darryl D Holm(Postdoc mentor), Department of Mathematics, Imperial College London, UK
Email: d.holm@imperial.ac.uk Ph: +44 2075948531